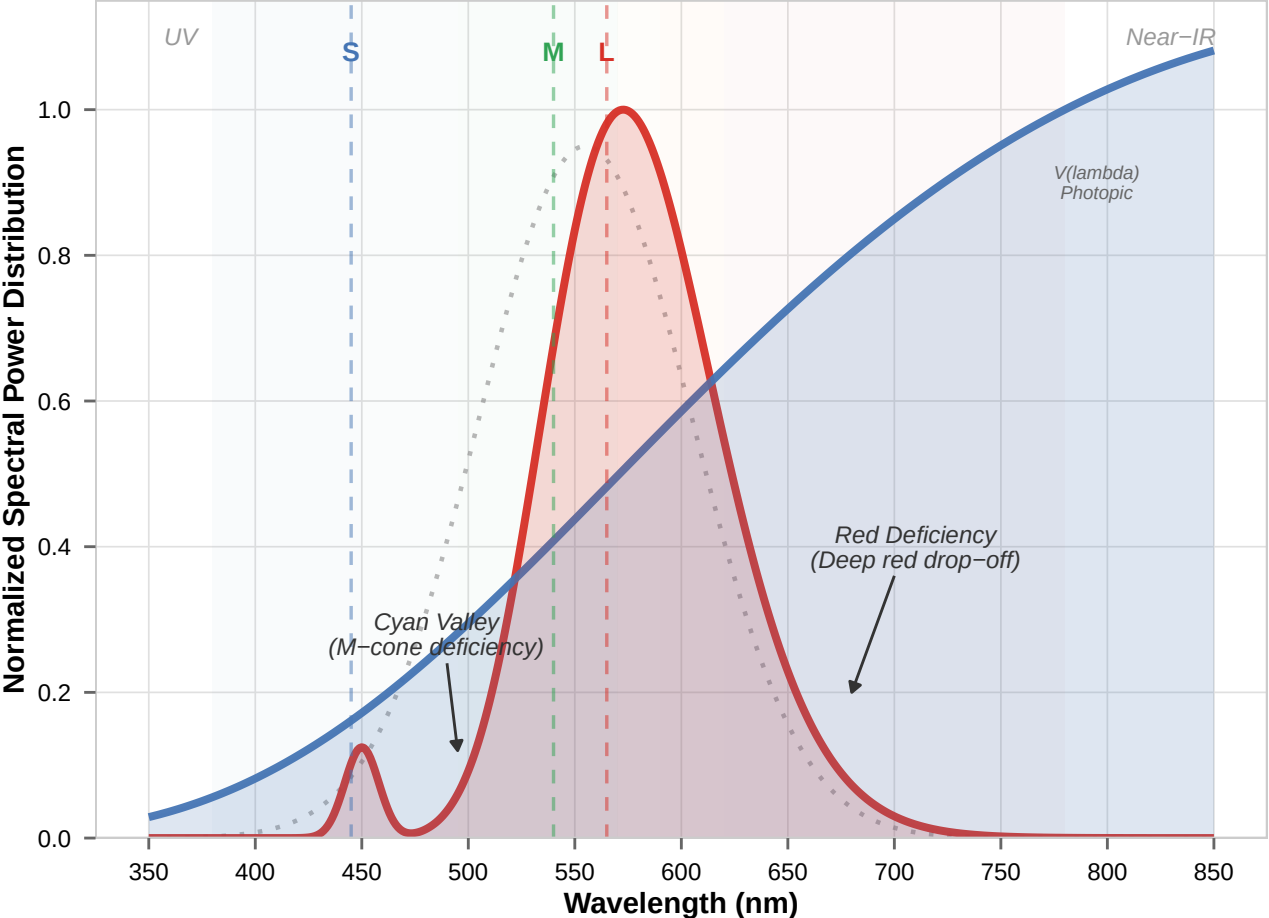


Figure 1.1: Spectral Power Distribution Comparison

Continuous (Tungsten) vs. Discontinuous (White LED) Light Sources

Tungsten (3000K Blackbody) White LED (InGaN + YAG:Ce)



Tungsten modeled using Planck's law at 3000K (smooth continuous spectrum). LED modeled as blue InGaN pump (450nm) + YAG:Ce (broad red background). Vertical dashed lines indicate L/M/S cone peak sensitivities. Dotted curve shows CIE 1931 photopic luminosity function $V(\lambda)$. Normalization: Each spectrum scaled to unity at its peak within visible range (380–780nm).